Document Management System

Sultanate of Oman
Ministry of Health
The Royal Hospital
Department of Child Health

Code: CH-PICU-GUD-4-Vers.1.0

Effective Date: 21/12/2022
Target Review Date: 21/12/2024

Title: Policy for the types and the indications for Central Venous Catheters in Paediatircs

ThisPolicy aims at assisting the choice of the different types of central venouscatheters in Children.It also describes the procedure for requesting forcentral venous catheters for pediatric patients.

What is A Central Venous Catheter?
- Central Venous Catheters (CVC) are different types of cathetersthat are inserted (percutaneously or surgically) in to a central vein (e.g SVC,IVC). They allow administration of intravenous therapies and blood sampling.The appropriate tip position for a central venous catheter is distal SVC orSVC/Right atrium (RA) junction using upper limbs veins. These catheters are insertedunder complete aseptic technique with continuous monitoring of vitals duringthe procedure.

Types of Central Venous Access (Catheters):
1. Short-term CVC (Temporary venous catheters)
2. Long-term CVC (PICC, Tunneled CVC, Indwelling venous Ports)

Temporary CVC:
These are catheters that are inserted in the neck or groin in the setting ofacute indications to provide treatment for a brief duration (Replacement with apermanent device should occur within 5-10 days when possible if permanentaccess is needed).

PeripherallyInserted Central Catheter (PICC):
These are catheters that are inserted in a peripheral vein (usually basilic,brachial or cephalic veins) and terminates at superior vena cava (SVC) or thejunction of SVC and right atrium. They are recommended when continuous venousaccess is required for a period from approximately 10 days until 6 months.

Tunneled CVC:
These are cuffed catheters that are inserted in a central vein and tunneledunder the skin from a distance with its tip terminating at SVC or SVC/RAjunction. They are used when a long-term continuous venous access is required,usually more than 6 months.
The insertion involve use of 2 incisions; one at the vein entrance site and theother at the catheter exist site (usually chest wall). A tunnel is made betweenthose 2 sites where fibrous sheath forms around the catheter few weeksfrominsertion.

Neck veins including internaljugular and subclavian veins are used for long term catheters.

Subcutaneous Venous Port:
These are subcutaneously implanted venousaccess devices which consist of a catheter inserted in IJV and implanted portunder the skin. These catheters are best used when intermittent chronic venousaccess is required. The Port is accessed as needed by trained staff for administrationof drugs or blood products.

The choice oflong term catheters is tailored according to:

- Ageand weight of the child.
- Typeof treatment required (daily or infrequent, one or multiple lumens).
- Durationof treatment.

Please consult Interventional Radiologyteam “IR” about the right choice for long term vascular access.

The available CVCs for use in children in the Royal hospital:

Temporary CVC:

- 4 Frtriple lumen
- 5 Frdouble or triple-lumen and 5.5 triple lumen

PICC:

- 1.9 Fr single-lumen
- 2.6 Fr double-lumen
- 3Fr single-lumen
- 4 Fr single or double-lumen

Tunneled CVC:

- 4.2Fr single-lumen (Hickman’s catheters-Bard)
- Othersize/choices will be available soon.

PortA Catheters:

- 5 Frsingle-lumen
- 5.5Fr single-lumen

- 6.5Fr single-lumen

Suggestions forchoice of CVCs based on treatment duration:

Temporary CVC: short treatment 5-10 days.

PICC: continuous daily treatment duration of 10 days - 6 months.

Tunneled CVC: continuous daily treatment > 6 months.

PortA Catheters: intermittent long-term treatment.

Indications forCVCs Insertion:

1. 1.Blood products
2. 2.Parenteralnutrition/lipids
3. 3. Antibiotics
4. 4.Dialysis/apheresis
5. 5.Chemotherapy
6. 6.Chelationtherapy
7. 7.Fluids/electrolytes
8. 8.Systemicanticoagulation/thrombolysis
9. 9.Delivery ofcaustic agents (e.g., Ca 2+)
10. 10. Other medications

Contra-indicationsfor CVC Insertion:

1. 1.Uncontrolledcoagulopathy:
- In children who requires these catheters emergently, their coagulopathy canbe optimized with the use of FFP, cryoprecipitate and platelets (Discussion with IR team is required)

2. 2. Bacteremiaand septicemia:
- It is recommended to treat underlying sepsis before insertion of long termtunneled catheters and ports for at least 48 hours.

Complications:

1. Bleeding.

2. Catheter infections. Rate of PICCsepsis=tunneled CVC in adults (1-3%).

3. Pneumothorax, hemothorax.

4. Injury to great vessels.

5. Venous thrombosis.

6. Catheter maintenance issues:

- Occlusion, movement, fracture and leaking of catheters.
- Fragmented catheters are retrievedfrom the heart and pulmonary vessels.
- Signs include swelling, pain at thesite and leakage around catheter.

7. Cardiac arrhythmias.

Clinicaldetails required prior to insertions of CVC:

- - Age and weight of the patient.
- - Current medications.
- - Purpose of the required CVC and duration oftreatment.
- - Previous CVC insertions.
- - Anticoagulation medications “if present”.
- - Status of CBC and coagulation profile.

Preparation forPercutaneous Insertions of CVCs:

- * Theseprocedures are usually done on Thursdays.
- * ContactIR team to schedule the child for the procedure.
- * Thechild needs to be reviewed by Anesthesia team.
- * Stopanticoagulation medications, if any “e.g. heparin/aspirin” according tohospital protocol prior to surgery.
- * Ensurethe child has recent laboratory investigations:
- * Thechild needs to remain fasting 6-8 hours prior to the procedure or as indicatedby the anesthesia team.

Blood Investigations required prior to the procedure:

- CompleteBlood Count (CBC): hemoglobin and platelet count (Platelets >100 or >75 on “case-basis”).
- Coagulation profile:
- Activated Prothrombin Time (APTT <35 sec).
- International Normalized Ratio (INR< 1.2).
- Cross match (only major procedures).

Removalof the CVC:

Ifthe catheter is not required any more or needs removal due to emergencyreasons, removal is done with/without local anesthesia for PICC/ temporary CVCand under General Anesthesia for tunneled CVC or PortA catheters. Cuffed PICCsneed to be removed by IR team.

References

Title of book/ journal/ articles/ Website	**Author**	**Year of publication**	
1.Pediatric interventional venous access.	Peter G.Chait, Michael Temple	**2002**	
2. Peripherally inserted central catheters.	Fabienne C.Bourgeois, Paula Lamagna	**2011**	**Page**
3. Central Venous access and related interventions. Pediatric Interventional Radiology, Springer	Temple, Michael. Marshalleck, Fracncis E	**2014**	

Written By: Dr.Yahya Al Brashdi / Dr.Kholoud Al Mukhaini
Checked By: PICU Team
Authorized by: Nashat Fuad Al-Sukaiti

'Technology Department'